

Santhosh Narayanan Baburaman

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EDUCATION

M.S. Analytics, University of Southern California (GPA: 3.92/4.0) Jan 2025 – Dec 2026

Coursework: Predictive Analytics, Optimization Methods, Cloud Analytics, Data Engineering

B.Tech. Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024

Publication: Co-authored an IEEE conference paper on predictive analytics for transportation, modeling travel time, fare, and traffic speed across 1M+ NYC taxi trip records

EXPERIENCE

Data Analyst Intern | Philadelphia Department of Aviation Jun 2026 – Aug 2026

- Designed the department's first structured reporting and data-verification layer across 2 revenue streams covering ~30% of airport operating revenue, replacing ~60 hours/month of manual review
- Engineered a Python → Parquet → Alteryx → Power BI pipeline over 800K+ rows of billing data spanning 4 fiscal years, replacing manual Excel review and surfacing anomalies the prior process missed
- Modeled a seasonal Z-score anomaly detector with peer-deviation and Benford's Law checks, and integrated ground transportation data with airline volumes to compute per-mode passenger capture rates

Founder & President | ISE Analytics Student Association (IASA), USC Feb 2025 – Present

- Founded USC's first analytics student organization, hosting 12+ events for 500+ students and 20+ professionals

PROJECTS

Healthcare Readmissions Prediction (GCP Vertex AI, Kubeflow, Scikit-learn)

- Owned data preparation for a 30-day hospital readmission model on 8,038 patient records with a 17% positive class
- Removed statistical outliers (z-score, modified z-score), engineered 3 clinical features, and applied SMOTE to the training split only; shipped preprocessing as versioned joblib artifacts
- Deployed the selected gradient boosting model (AUPRC 0.86) through an 8-component Kubeflow pipeline on Vertex AI with model registry and endpoint serving

Automated Sales Analytics Copilot (Python, PostgreSQL, Claude API, Streamlit) 

- Built an agentic LLM system with multi-step tool calling, text-to-SQL, and a 6-tool structured reasoning layer over a live PostgreSQL database
- Developed ARIMA and Prophet forecasting with 80% confidence intervals and a 4-rule anomaly engine (z-score outliers, MoM trend reversals, margin compression)
- Shipped a full-stack app on Streamlit with nightly GitHub Actions inference and automated Slack and email alerting

eBay ML Challenge — Transformer NER (Python, PyTorch, NLP) 

- Fine-tuned DistilBERT for Named Entity Recognition, extracting brand, model, and color attributes from eBay listings across 30K+ samples
- Trained category-specific models with early stopping and macro-F1 tracking; automated token alignment and label mapping for reproducible inference

RL Trading & Portfolio Optimization (Python, Reinforcement Learning) 

- Formulated a finite-horizon Markov Decision Process with defined state, action, and reward spaces for trade execution decisions
- Implemented tabular Q-learning with Softmax exploration to learn execution and portfolio allocation policies, evaluated on cumulative reward and liquidation metrics

IPL 2026 Playoff Tracker (Python, Monte Carlo, Claude API) 

- Built a 5,000-run Monte Carlo simulation estimating playoff qualification odds for all 10 teams, using a logistic win-probability model weighted on points, net run rate, and form

SKILLS

- **Languages & ML:** Python, SQL, R, PyTorch, Scikit-learn, Pandas, NLP, Transformer Models, Reinforcement Learning (Q-learning), Predictive Modeling, Anomaly Detection
- **Data & Cloud:** PostgreSQL, Snowflake, Redshift, Alteryx, Power BI, ARIMA, Prophet, GCP (Vertex AI, Kubeflow), Azure, LLM APIs, Git